ASAP CRN Cloud PMDBS bulk RNAseq Collection

README - currently release 1.0.0, 10.5281/14373344

Overview

The ASAP CRN Cloud is pleased to announce the inaugural (v1.0.0) **PMDBS bulk RNAseq Collection** with the v2.0.0 ASAP CRN Cloud release.

ASAP Teams: Team Lee, Team Hardy, Team Wood
Lead Principal Investigators: Michael Lee, John Hardy, Nicholas Wood,
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See CRN Cloud Authorship List for the full list of contributing investigators/researchers by dataset.
ASAP CRN GitHub: github.com/ASAP-CRN
Data Release Date: 9/11/24
ASAP CRN Cloud Release Number: v2.0.0
Release DOI: 10.5281/zenodo.14270014
Curated Data File Manifest: ASAP CRN Cloud File Manifest - v3.0

PMDBS bulk RNAseq Collection

The raw bulk RNAseq data has been processed and harmonized to create a harmonized database of gene expression data. The platformed data is summarized below as Upstream, Downstream and Cohort analysis summaries below. Additional explanation for the analysis pipeline employed for processing the platformed data can be found at the ASAP-CRN pmdbs-bulk-rnaseq-wf github repo.

Collection details

PMDBS bulk RNAseq Workflow Release - v1.0.0

- Release date: December 3, 2024
- GitHub release link

- Pipeline version: v1.0.0

Collection Datasets list

- `team-hardy-pmdbs-bulk-rnaseq`
- `team-lee-pmdbs-bulk-rnaseq-mfg`
- `Team-wood-pmdbs-bulk-rnaseq`
- `cohort-pmdbs-bulk-rnaseq.` This *cohort* dataset includes samples integrated from two of the contributed bulk RNAseq datasets:
  - `team-hardy-pmdbs-bulk-rnaseq`
  - `team-wood-pmdbs-bulk-rnaseq`

Curated Data

The curated data can be categorized as "*upstream*" and "*downstream*", and "*cohort analysis*". They are further segregated by mode: *"mapping mode"* and *"alignment mode"*.

Upstream.
The bulk RNAseq pipeline begins with QC and trimming of raw fastq files to ensure high-quality input for downstream processing. Following these steps, users can select one or both of the available processing modes: alignment mode and/or mapping mode. In alignment mode, the trimmed fastq files are aligned to a reference genome using STAR, producing BAM files with aligned reads. These BAM files are then processed using Salmon (alignment mode) to quantify gene expression, generating output such as transcript counts or TPM values. In contrast, mapping mode bypasses the alignment step and directly quantifies gene expression from the trimmed fastq files using Salmon (mapping mode).

Downstream
Differential analysis is performed with PyDeseq2 to identify statistically significant differentially expressed genes between Parkinson's Disease and Control samples. The filtering cutoffs were chosen to be:

- Adjusted p-value < 0.05
- Absolute log2 fold change > 1

Cohort analysis
A cohort analysis is made by looking across all the gene expression data to find modes of expression across genes between our experimental groups.

Raw Data

The *raw* data refers to the fastq files transferred by each ASAP CRN Team. These data are available with the caveat that they are stored in “requester pays” buckets on the google cloud platform. A google cloud billing account will need to be registered to access these data. More information is available HERE. gs://`asap-raw-<team_name>-<source>-<dataset_name>`

Metadata & Data Dictionary

Data Dictionary: ASAP CRN Cloud Data-Dictionary - v3.0

The ASAP CRN Common Data Elements (CDE - v3.0) outlines the variables which are harmonized across the ASAP CRN. These metadata enable the harmonization of the data submitted by multiple teams. The metadata consists of five tables, which are described below, and also detailed as Data Dictionaries for each.

Metadata refers to descriptive information about the overall study, individual samples, all protocols, and references to processed and raw data file names. Metadata was aggregated using the table-based submission for each team's dataset. The tables - STUDY, PROTOCOL, SUBJECT, SAMPLE, DATA ,etc. - can be thought of as spreadsheets, and each table is prepared as simple .csv files which constitute the tables outlined in the ASAP CRN CDE (final sheet). These have been aggregated across submissions and unique ASAP IDs generated for each dataset, team, subject and sample. I.e. ASAP_dataset_id, ASAP_team_id, ASAP_subject_id, and ASAP_sample_id.

The metadata can be browsed on ASAP CRN Cloud *Explorer* and exposed in Verily Workbench.

This document.